Ethics in Information Technology, Fourth Edition

Computer and Internet Crime
Chapter 3

Objectives

- As you read this chapter, consider the following questions:
  - What key trade-offs and ethical issues are associated with the safeguarding of data and information systems?
  - Why has there been a dramatic increase in the number of computer-related security incidents in recent years?
  - What are the most common types of computer security attacks?

Objectives (cont'd.)

- Who are the primary perpetrators of computer crime, and what are their objectives?

IT Security Incidents: A Major Concern

- Security of information technology is of utmost importance
  - Safeguard:
    - Confidential business data
    - Private customer and employee data
  - Protect against malicious acts of theft or disruption
  - Balance against other business needs and issues
- Number of IT-related security incidents is increasing around the world

Why Computer Incidents Are So Prevalent

- Increasing complexity increases vulnerability
  - Computing environment is enormously complex
    - Continues to increase in complexity
    - Number of entry points expands continuously
    - Cloud computing and virtualization software
- Higher computer user expectations
  - Computer help desks under intense pressure
    - Forget to verify users' IDs or check authorizations
- Computer users share login IDs and passwords

Why Computer Incidents Are So Prevalent (cont'd.)

- Expanding/changing systems equal new risks
  - Network era
    - Personal computers connect to networks with millions of other computers
    - All capable of sharing information
  - Information technology
    - Is everywhere
    - Necessary tool for organizations to achieve goals
    - Increasingly difficult to match pace of technological change for risk assessment

Why Computer Incidents Are So Prevalent (cont'd.)

- Increased reliance on commercial software with known vulnerabilities
  - Exploit
    - Attack on information system
    - Takes advantage of system vulnerability
    - Due to poor system design or implementation
  - Patch
    - "Fix" to eliminate the problem
    - Users are responsible for obtaining and installing
    - Delays expose users to security breaches

Why Computer Incidents Are So Prevalent (cont'd.)

- Zero-day attack
  - Before a vulnerability is discovered or fixed
- U.S. companies rely on commercial software with known vulnerabilities

Types of Exploits

- Computers as well as smartphones can be target
- Types of attacks
  - Virus
  - Worm
  - Trojan horse
  - Distributed denial of service
  - Rootkit
  - Spam
  - Phishing (spear-phishing, smishing, and vishing)

Viruses

- Pieces of programming code
- Usually disguised as something else
- Cause unexpected and undesirable behavior
- Often attached to files
- Deliver a "payload"
- Spread by actions of the "infected" computer user
    - Infected e-mail document attachments
    - Downloads of infected programs
    - Visits to infected Web sites

Worms

- Harmful programs
  - Reside in active memory of a computer
  - Duplicate themselves
- Can propagate without human intervention
- Negative impact of worm attack
  - Lost data and programs
  - Lost productivity
  - Additional effort for IT workers

Trojan Horses

- Malicious code hidden inside seemingly harmless programs
- Users are tricked into installing them
- Delivered via email attachment, downloaded from a Web site, or contracted via a removable media device
  - Some pirated copies of this software contain a Trojan horse
- Logic bomb
  - Executes when triggered by certain event

Distributed Denial-of-Service (DDoS) Attacks

- Malicious hacker takes over computers on the Internet and causes them to flood a target site with demands for data and other small tasks
  - The computers that are taken over are called zombies
  - Botnet is a very large group of such computers
- Does not involve a break-in at the target computer
  - Target machine is busy responding to a stream of automated requests
  - Legitimate users cannot access target machine

Rootkits

- Set of programs that enables its user to gain administrator-level access to a computer without the end user’s consent or knowledge
- Attacker can gain full control of the system and even obscure the presence of the rootkit
- Fundamental problem in detecting a rootkit is that the operating system currently running cannot be trusted to provide valid test results

Rootkits

- symptoms of rootkit infections:
  - The computer locks up or fails to respond to input from the keyboard or mouse.
  - The screen saver changes without any action on the part of the user.
  - The taskbar disappears.
  - Network activities function extremely slowly.

Spam

- Abuse of email systems to send unsolicited email to large numbers of people
  - Low-cost commercial advertising for questionable products
  - Method of marketing also used by many legitimate organizations
- Controlling the Assault of Non-Solicited Pornography and Marketing (CAN-SPAM) Act
  - Legal to spam if basic requirements are met

Spam (cont'd.)

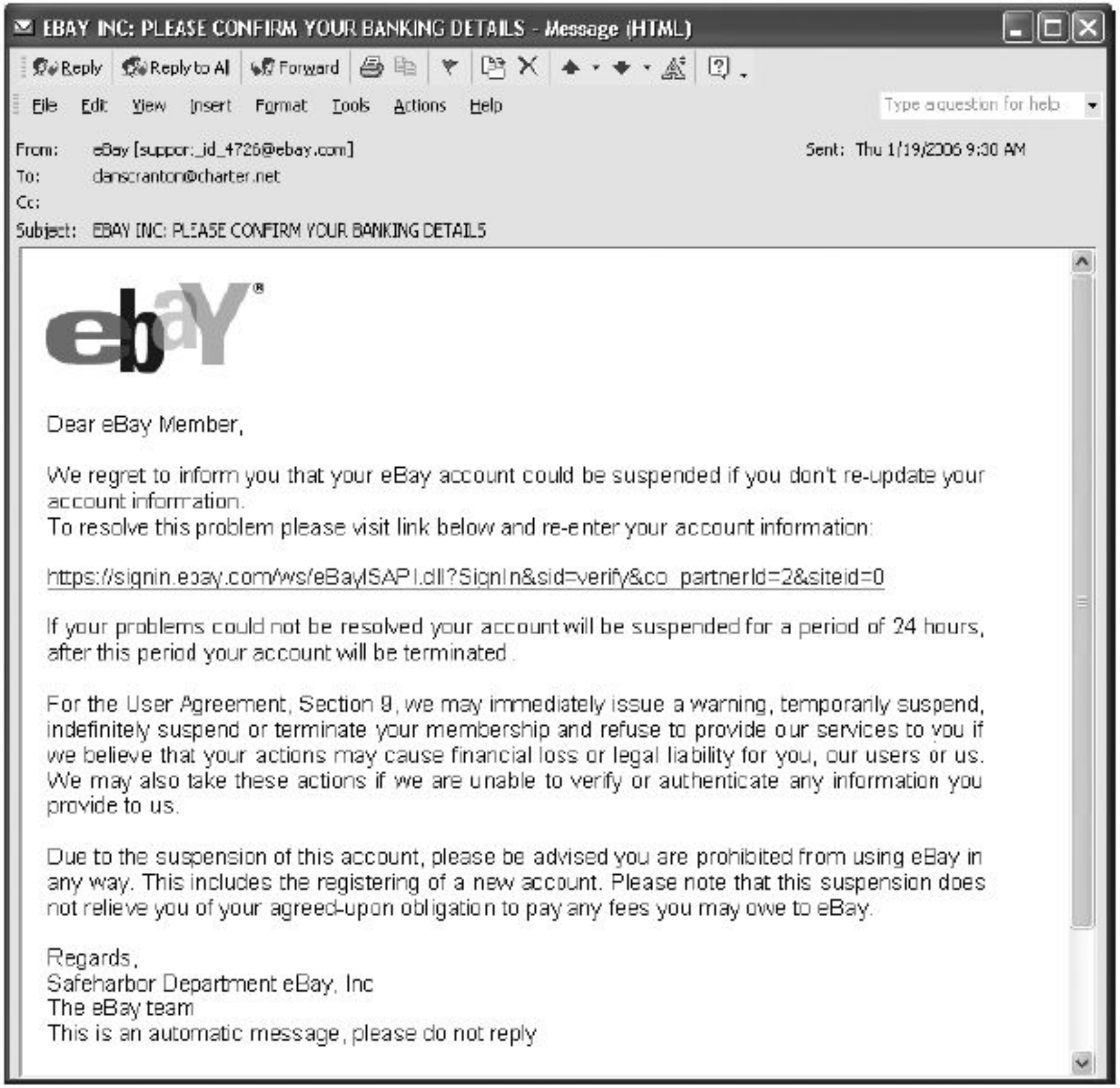
- Completely Automated Public Turing Test to Tell Computers and Humans Apart (CAPTCHA)
  – Software generates tests that humans can pass but computer programs cannot

Spam (cont'd.)

- For nearly five years, Edward Davidson ran a spamming, he managed a large network of computers that sent hundreds of thousands of spam e-mails, promoted the sale of watches, perfume, and other items for nearly two dozen companies. Davidson and his subcontractors sent e-mail messages with header information that concealed the actual sender from the recipient of the e-mail—a violation of the federal CAN-SPAM Act. In April 2008, Edward Davidson was sentenced to serve 21 months in federal prison for violation of the CAN-SPAM Act. He was also ordered to pay $714,139 in restitution to the IRS for taxes on income from the operation that he failed to report.

Phishing

- Act of using email fraudulently to try to get the recipient to reveal personal data
- Legitimate-looking emails lead users to counterfeit Web sites
- Spear-phishing
  - Fraudulent emails to an organization’s employees
- Smishing
  - Phishing via text messages
- Vishing
  - Phishing via voice mail messages



EBAY INC: PLEASE CONFIRM YOUR BANKING DETAILS - Message (HTML)

Reply | Reply to All | Forward

File Edit View Insert Format Tools Actions Help

Type a question for help

From: eBay [suppor_id_4726@ebay.com]

Sent: Thu 1/19/2006 9:30 AM

To: danscranton@charter.net

Cc:

Subject: EBAY INC: PLEASE CONFIRM YOUR BANKING DETAILS

Dear eBay Member,

We regret to inform you that your eBay account could be suspended if you don't re-update your account information.
To resolve this problem please visit link below and re-enter your account information:

https://signin.ebay.com/ws/eBayISAPI.dll?SignIn&sid=verify&co_partnerId=2&siteid=0

If your problems could not be resolved your account will be suspended for a period of 24 hours, after this period your account will be terminated.

For the User Agreement, Section 9, we may immediately issue a warning, temporarily suspend, indefinitely suspend or terminate your membership and refuse to provide our services to you if we believe that your actions may cause financial loss or legal liability for you, our users or us. We may also take these actions if we are unable to verify or authenticate any information you provide to us.

Due to the suspension of this account, please be advised you are prohibited from using eBay in any way. This includes the registering of a new account. Please note that this suspension does not relieve you of your agreed-upon obligation to pay any fees you may owe to eBay.

Regards,
Safeharbor Department eBay, Inc
The eBay team
This is an automatic message, please do not reply

FIGURE 3-3 Example of phishing

Source Line: Course Technology/Cengage Learning.

Types of Perpetrators

- Perpetrators include:
  - Thrill seekers wanting a challenge
  - Common criminals looking for financial gain
  - Industrial spies trying to gain an advantage
  - Terrorists seeking to cause destruction
- Different objectives and access to varying resources
- Willing to take different levels of risk to accomplish an objective

Types of Perpetrators (cont'd.)

TABLE 3-4 Classifying perpetrators of computer crime

Type of perpetrator	Typical motives
Hacker	Test limits of system and/or gain publicity
Cracker	Cause problems, steal data, and corrupt systems
Malicious insider	Gain financially and/or disrupt company's information systems and business operations
Industrial spy	Capture trade secrets and gain competitive advantage
Cybercriminal	Gain financially
Hacktivist	Promote political ideology
Cyberterrorist	Destroy infrastructure components of financial institutions, utilities, and emergency response units

Source Line: Course Technology/Cengage Learning.

Hackers and Crackers

- Hackers
  - Test limitations of systems out of intellectual curiosity
    - Some smart and talented
    - Others inept; termed “lamers” or “script kiddies”
- Crackers
  - Cracking is a form of hacking
  - Clearly criminal activity

Malicious Insiders

- Major security concern for companies
- Fraud within an organization is usually due to weaknesses in internal control procedures
- Collusion
  - Cooperation between an employee and an outsider
- Insiders are not necessarily employees
  - Can also be consultants and contractors
- Extremely difficult to detect or stop
  - Authorized to access the very systems they abuse
- Negligent insiders have potential to cause damage

Industrial Spies

- Use illegal means to obtain trade secrets from competitors
- Trade secrets are protected by the Economic Espionage Act of 1996
- Competitive intelligence
  - Uses legal techniques
  - Gathers information available to the public
- Industrial espionage
  - Uses illegal means
  - Obtains information not available to the public

Cybercriminals

- Hack into corporate computers to steal
- Engage in all forms of computer fraud
- Chargebacks are disputed transactions
- Loss of customer trust has more impact than fraud

Hacktivists and Cyberterrorists

- Hacktivism
  - Hacking to achieve a political or social goal
- Cyberterrorist
  - Attacks computers or networks in an attempt to intimidate or coerce a government in order to advance certain political or social objectives
  - Seeks to cause harm rather than gather information
  - Uses techniques that destroy or disrupt services

Federal Laws for Prosecuting

TABLE 3-5 Federal laws that address computer crime

Federal law	Subject area
USA Patriot Act	Defines cyberterrorism and penalties
Identity Theft and Assumption Deterrence Act (U.S. Code Title 18, Section 1028)	Makes identity theft a Federal crime with penalties up to 15 years imprisonment and a maximum fine of $250,000
Fraud and Related Activity in Connection with Access Devices Statute (U.S. Code Title 18, Section 1029)	False claims regarding unauthorized use of credit cards
Computer Fraud and Abuse Act (U.S. Code Title 18, Section 1030)	Fraud and related activities in association with computers: • Accessing a computer without authorization or exceeding authorized access • Transmitting a program, code, or command that causes harm to a computer • Trafficking of computer passwords • Threatening to cause damage to a protected computer

Summary

- Ethical decisions in determining which information systems and data most need protection
- Most common computer exploits
  - Viruses
  - Worms
  - Trojan horses
  - Distributed denial-of-service attacks
  - Rootkits
  - Spam
  - Phishing, spear-fishing, smishing, vishing

Summary (cont'd.)

- Perpetrators include:
  - Hackers
  - Crackers
  - Malicious insider
  - Industrial spies
  - Cybercriminals
  - Hacktivist
  - Cyberterrorists

Summary (cont'd.)

- Must implement multilayer process for managing security vulnerabilities, including:
  - Assessment of threats
  - Identifying actions to address vulnerabilities
  - User education
- IT must lead the effort to implement:
  - Security policies and procedures
  - Hardware and software to prevent security breaches
- Computer forensics is key to fighting computer crime in a court of law